

AMAZON PRICE TRACKER PRO

Proof of Concept — Project Summary

A browser extension that watches Amazon prices for you,
and lets you know the moment a good deal shows up.

What Is This Project?

Amazon Price Tracker Pro is a small Chrome extension. You give it a product on Amazon and a price you'd be happy to pay. It quietly checks that product every so often, and as soon as the price drops to your target, it sends you a notification — click it, and you're taken straight to the product page to buy.

This document explains, in plain language, what has been built so far, how it works, and how to run and publish it.

In short: It's a personal shopping assistant that watches prices while you don't have to.

How It's Built

The project has two halves that talk to each other:

- **Chrome Extension (the app you see):** Built using Chrome's modern Manifest V3 standard. This is what sits in your browser toolbar.
- **Backend Server (the brain behind the scenes):** A FastAPI (Python) server connected to a PostgreSQL database, which stores accounts, tracked products, and alert history.

These two pieces communicate over the internet: the extension sends and receives data, and the server saves it and keeps track of everything long-term.

What It Can Already Do

Here is a quick-glance summary of every feature built so far:

Feature	Status	What it does
Sign in & accounts	Done	Your own login, separate from your Amazon account
Track a product	Done	Save any Amazon product with one click
Automatic price checks	Done	Checks prices in the background, no action needed
Price-drop alerts	Done	Desktop notification the moment a target price is hit
One-click buy	Done	Clicking the alert opens the product, ready to buy
Alert history log	Done	A timeline of every alert that's ever fired

1. Signing In — Two Separate Logins, Working Together

There are two logins involved, and it's worth being clear they don't mix:

- **Your Amazon login stays exactly as it is.** You log into Amazon normally in Chrome, and the extension simply works alongside that — it never touches your Amazon username or password.
- **A separate app login.** To use the extension itself (save products, see your dashboard), you create a lightweight account secured with an industry-standard token system (JWT), so only you can see your own tracked items.

2. Saving a Product to Track

When you're on a product page on Amazon, the extension quietly reads the page and picks up three things: the product's name, its web address, and its current price.

From there, using it is simple:

1. Click the extension icon in your toolbar.
2. Optionally, type in a target price — the price you're waiting for.
3. Click "Track This Product."

The product is saved to your account, and it now shows up on your dashboard — a simple list of everything you're currently tracking.

3. Checking Prices Automatically

This is the part that runs quietly in the background, without you having to do anything:

- The extension wakes itself up on a timer (currently every 1 minute, for testing — this would be much less frequent in a real, everyday version).
- Each time it wakes up, it looks at your tracked list, visits each product's page in the background, and reads off the current price and whether it's in stock.

Good to know: The 1-minute interval is only for testing this proof of concept quickly. A real version would check far less often — for example, once every few hours — to be respectful of Amazon's website.

4. Getting Notified of a Price Drop

If a background check finds that a product's price has fallen to, or below, your target price, the extension immediately sends a standard desktop notification. It clearly shows the product name and the new, lower price, so you don't miss it.

5. One Click to Buy

Notifications aren't just an alert — they're actionable. Clicking on the notification instantly opens a new browser tab, taking you straight to that product's Amazon page, ready for you to purchase without any extra searching.

6. A History of Every Alert

Every time an alert fires, it's also recorded to the backend server. Inside the extension, a dedicated "Logs" tab shows a simple, chronological timeline of every price alert you've ever received — useful for looking back at deals you caught.

Important Things to Know

- **How prices are read:** The extension reads prices directly off the Amazon page's content. If Amazon changes the way its pages are laid out, this reading logic will occasionally need small updates to keep working.
- **Respecting Amazon's systems:** Because checks happen through your own browser and your own internet connection — the same way as if you visited the page yourself — this is gentler on Amazon's systems than a server doing it on your behalf. That said, the extension does not and will not try to bypass or trick any of Amazon's security or anti-bot systems.
